No.	Status	Date
QSCP-NDT-001 Rev 4	Issued	2025-06-16

Site NDE Procedure - Ultrasonic Thickness Measurement, ATU-100

1.0 Scope and Application

This procedure establishes the requirements for ultrasonic thickness measurement within the ATU-100, superseding QSCP-NDT-001 Rev 3. It incorporates updated requirements for PAUT corrosion mapping at rich-amine fittings where readings fall below the alert level, ensuring high-resolution data for localized degradation. The scope includes all piping circuits identified in QSCP-IP-ATU100 Rev 4, specifically targeting rich-amine and sour gas services.

All measurements must be performed using the ATU-100 gauge or equivalent digital ultrasonic equipment. For A312 TP316 stainless steel lines, the instrument must be calibrated to the specific stainless-steel velocity setting. This revision ensures that CML-id validation is performed during data export to maintain database integrity and alignment with the corporate inspection management system.

2.0 Personnel Qualifications and Responsibilities

NDE personnel performing thickness surveys must be certified to at least Level II in accordance with the SNT-TC-1A written practice or CAN/CGSB-48.9712. Technicians are responsible for verifying CML-id formats against the master list and ensuring that any reading below the alert level is supported by a photo of the gauge display. Jennifer Walsh, as Inspection Coordinator, is responsible for the technical oversight of the NDE program and the verification of all field data.

Robert Hendricks, as Unit Engineer, is responsible for reviewing and approving the final inspection results and any subsequent repair recommendations. The inspection team must ensure that follow-up dates for inaccessible locations, such as 6-LA-103-08, are tracked as noted in LNDE-RPT-20241025-031-R2. Any deviation from this procedure requires written authorization from the Inspection Department.

3.0 Inspection Intervals and Site Rules

Inspection intervals are governed by the RBI strategy outlined in QSCP-IP-ATU100 Rev 4, which mandates a 3-year interval for Class 1 sour gas and rich amine lines. Remaining life calculations utilize the larger of the long-term or short-term corrosion rates, with the short-term rate requiring a minimum 3-year interval between readings. The next inspection must be scheduled at half the remaining life or the class maximum, whichever is sooner.

Specific re-measurement dates are established for critical assets: 4-RA-101 is scheduled for Aug 2025, 6-RA-103-11 for Feb 2026, and 2-RF-102 for Nov 2025. Retirement thickness is defined as the line t-min; any reading recorded below this value must be immediately reported as an NCR. For insulated lines within the CUI range, profile RT or insulation windows are required every 5 years.

4.0 Data Management and Reporting

All thickness data must be exported digitally with validated CML-id formats to prevent synchronization errors. Reports must include a cross-reference to PAUT results where corrosion mapping was triggered by low thickness readings. This revision addresses the corrections required in LNDE-RPT-20241025-031-R2 regarding Appendix B formatting and the inclusion of previously omitted follow-up dates.

Final reports shall be reviewed against the historical data provided in QSCP-NDT-001 Rev 3 to ensure consistency in corrosion rate trends. The Inspection Coordinator, Jennifer Walsh, will validate that all Class 2 amine and treated gas lines adhere to the 10-year thickness and 5-year external visual maximums. Robert Hendricks will provide final engineering sign-off on the integrity status of the ATU-100 piping circuits.

Sign-off	Name	Function	Date	Signature
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2025-06-16	_____
Approved	Robert Hendricks	Unit Engineer ATU-100, P.Eng.	2025-06-16	_____